3. SEQUENCE OPERATIONS AND SIGNAL ANALYSIS

Objective

This experiment is divided into four simulations:

1. Addition and Multiplication of Two Sequences
2. Energy and Power Calculation of Given Signals
3. Autocorrelation
4. Cross Correlation

Experiment 3.1

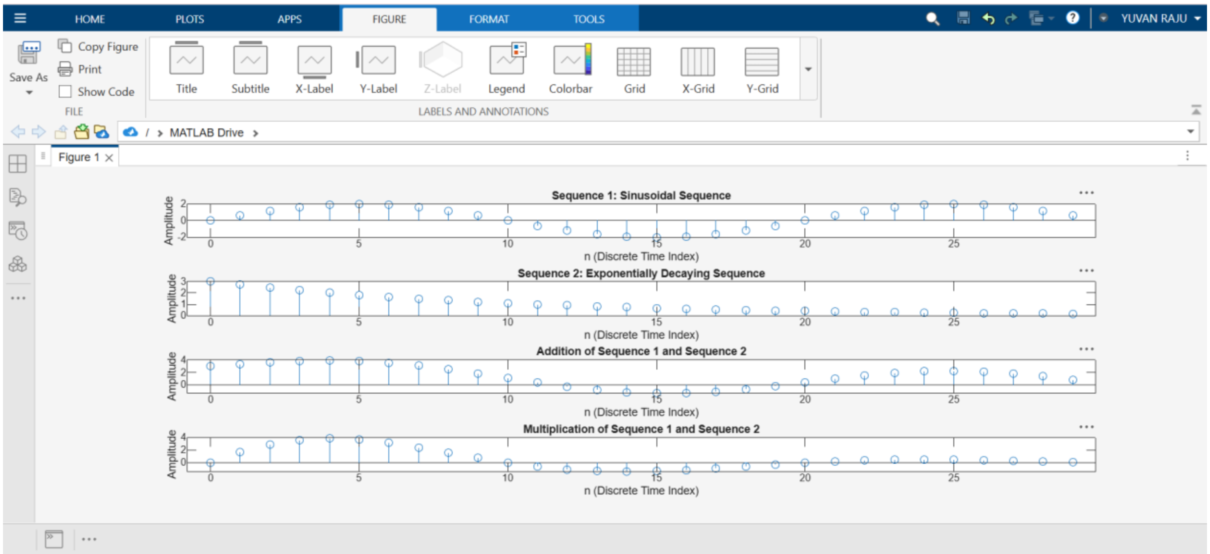
Addition and Multiplication of Two Sequences

Program:

```
// Test Case: Addition and Multiplication of Two Sequences
// Define parameters
N = 30; // Number of samples
n = 0:N-1; // Discrete time index vector
// Define Sequence 1: Sinusoidal sequence
f1 = 0.05; // Frequency in cycles/sample
A1 = 2; // Amplitude
x1_n = A1 * sin(2 * %pi * f1 * n);
// Define Sequence 2: Exponentially decaying sequence
a2 = 0.1; // Base of the exponential decay
A2 = 3; // Amplitude
x2_n = A2 * exp(-a2 * n);
// Addition of the two sequences
x_add = x1_n + x2_n;
// Multiplication of the two sequences
x_mult = x1_n .* x2_n;
// Plot the sequences and their results
// Plot Sequence 1
```

```
clf; // Clear the current figure
subplot(4,1,1);
stem(n, x1_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Sequence 1: Sinusoidal Sequence');
// Plot Sequence 2
subplot(4,1,2);
stem(n, x2_n);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Sequence 2: Exponentially Decaying Sequence');
// Plot the result of addition
subplot(4,1,3);
stem(n, x_add);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Addition of Sequence 1 and Sequence 2');
// Plot the result of multiplication
subplot(4,1,4);
stem(n, x_mult);
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Multiplication of Sequence 1 and Sequence 2');
```

OUTPUT



Experiment 3.3

Autocorrelation

Program:

```
// Test Case: Autocorrelation of a Rectangular Pulse Sequence

// Define parameters

N = 20; // Number of samples

x_n = [ones(1,5), zeros(1,N-5)]; // Rectangular pulse sequence with width 5

// Compute the autocorrelation

R_xx = correl(x_n, x_n);

// Create a lag vector for plotting

lags = -N+1:N-1;

// Plot the input sequence

clf; // Clear the current figure

subplot(2,1,1);

bar(0:N-1, x_n, 'b');

xlabel('n (Discrete Time Index)');

ylabel('Amplitude');
```

```
title('Input Sequence: Rectangular Pulse Sequence');

// Plot the autocorrelation sequence
subplot(2,1,2);
bar(lags, R_xx, 'r');
xlabel('Lag');
ylabel('Autocorrelation');
title('Autocorrelation of the Rectangular Pulse Sequence');
```

OUTPUT